Nabua National High School

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Research Plan

Working Title: Effect of Moringa *oleifera* Leaf Extract-Infused Hydrogels on Seed Germination of *Vigna Radiata* on a gel-based growing system

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CHAPTER 1
RESEARCH PROPOSAL

BACKGROUND OF THE STUDY

Seed germination is a crucial stage in the life cycle of plants, as it marks the beginning of their growth and development. It is a complex process that involves the activation of enzymes, the uptake of water and nutrients, and the emergence of the radicle, which eventually develops into the root system. One of the most critical factors affecting seed germination is water availability. Seeds require water to initiate metabolic processes and to activate enzymes that trigger germination process. However, soil-based systems are often challenged by poor water availability, soil degradation, and vulnerability to stress, prompting interest in different medium that support seedling growth (Iqbal, 2023).

In recent years, gel-based systems have gained increasing interest from researchers across many fields. Gel-based systems, particularly hydrogel, it functions as a storage of water where it is released slowly for efficient water use. Hydrogel polymers have a three-dimensional structure and exhibit hydrophilic properties, which create unique properties, such as high-water absorption capacity, flexibility, and biocompatibility. However, most of the existing research on hydrogels mainly focuses on incorporating inorganic fertilizers or nutrients into hydrogels, while fewer studies have examined the infusion of natural extracts into hydrogels (Muhammad et al., 2025).

Plant extracts, such as those extracted from moringa (*Moringa oleifera*) leaves, are known to contain bioactive compounds, including minerals, antioxidants, and plant growth promoting substances (Al Husnan & Alkahtani, 2016). Several studies have demonstrated the effectiveness of moringa leaf extracts in enhancing seed germination, seedling vigor, and overall plant growth. Separately, hydrogels have also been shown to improve water availability during germination. Despite this, research that infuses plant extracts into hydrogels remain limited in agricultural contexts.

Most studies on extract-infused hydrogels have been conducted in biomedical or material science fields, such as wound dressing and drug delivery, rather than in agricultural or seed germination contexts. As a result, there is insufficient understanding of how infusing plant extracts into hydrogels influences their moisture retention behavior and their effectiveness at the seed-hydrogel interface during germination. It is also unclear whether the combined use of plant extract and hydrogel offers additive or synergistic benefits compared to their individual application.

Given the accessibility of moringa leaves and the fast rate of germination of mung beans, a species commonly used as a model in seed germination studies due to its sensitivity to moisture availability, investigating the use of moringa leaf extract-infused hydrogels as a germination medium presents a practical and relevant research opportunity. This study aims to address the existing research gap by evaluating the effects of extract-infused hydrogels on seed germination parameters, contributing to the development of low-cost, sustainable, and student friendly cultivation strategies.

STATEMENT OF THE PROBLEM

Farming systems often suffer from inadequate water retention during seed germination and early plant growth, especially during dry seasons and El Niño events, leading to delayed germination and weak seedlings. Hydrogels can absorb and gradually release water due to their three-dimensional polymer structure, making them suitable for providing consistent moisture in gel-based growing systems (Muhammad et al., 2025).

Moringa *oleifera* leaf extract contains bioactive compounds that may enhance seed germination and early growth (Al Husnan & Alkahtani, 2016), yet most hydrogel studies focus on inorganic additives, while extract-infused hydrogels are mainly explored in biomedical applications. There is limited evidence on the effectiveness of moringa extract–infused hydrogels for seed germination in fast-germinating crops such as mung bean (*Vigna radiata*), which this study aims to address.

General Problem

The objective of this study entitled, Effect of Moringa (*Moringa Oleifera*) Leaf Extract-Infused Hydrogels on Seed Germination in a Gel-Based Growing System is to answer the question: How effective is a hydrogel infused with Moringa oleifera leaf extract in improving the germination and early growth of mung bean (*Vigna radiata*).

Specific Problem

Specifically, this study sought answer to the following questions:

1. What is the effect of Moringa oleifera leaf extract-infused hydrogel on mung bean seed germination in terms of:
    a. Germination Percentage (%) (G%)
    b. Mean Germination Time (MGT)
    c. Shoot Length (cm) (SL)
    d. Root Length (cm) (RL)
    e. Seedling Vigor Index (SVI)
2. Which of the treatment (0%, 5%, 15% moringa extract concentration) showed the highest germination percentage of mung beans?
3. Is there a significant difference in the seed germination in terms of G%, MGT, SL, RL, and SVI of *Vigna Radiata* in different concentration levels of Moringa oleifera extracts?

HYPOTHESIS

H0 – There is no significant difference in seed germination in terms of of G%, MGT, SL, RL, and SVI of *Vigna Radiata* among the treatments (CT1 – 0%, T1 – 5%, T2 – 15%).

H1 - There is a significant difference in seed germination in terms of G%, MGT, SL, RL, and SVI of *Vigna Radiata* among the treatments (CT1 – 0%, T1 – 5%, T2 – 15%).

SIGNIFICANCE OF THE STUDY

This study seeks to determine the effectiveness of Moringa oleifera leaf extract-infused hydrogels in gel-based growing systems and its potential contribution to sustainable plant cultivation, particularly during the seed germination stage.

Other Researcher/s: This study may serve as a foundation for future researchers exploring the use of hydrogels in gel-based or soilless growing systems. It may also encourage further agricultural research on the application of plant-based extracts, such as Moringa oleifera, as natural and sustainable alternatives for improving early plant growth.

Community and Environment: This study may contribute to improved crop production in rain-fed areas, soilless growing systems, and regions with irregular irrigation or limited water supply. By prompting efficient water use through hydrogels and natural plant extracts, the study supports environmentally sustainable practices and may help reduce water stress during early plant development.

Scopes and Delimitations: This study focuses on the use of hydrogels infused with moringa leaf extract in a gel-based growing system and their effect on the seed germination of *Vigna Radiata*. The experiment is limited to the germination stage only, and plant growth beyond early seedling growth is not included.

This study considers germination percentage as the primary parameter evaluating the effectiveness of the treatments. Other growth indicators such as, leaf development, and yield are not covered. The extraction of moringa leaves is limited to the aqueous extraction method due to its feasibility to students.

DEFINITION OF TERMS

Aqueous Extraction

(c) uses water as solvent to pull water-soluble compounds from solid materials (Al Husnan & Alkahtani, 2016).

(o) The extraction technique that is to extract moringa leaf extracts in this study.

Extract-Infused Hydrogel

(c) They incorporate agents such as natural extracts and metal/metal oxide nanoparticles to boost antimicrobial properties (Hossainpour et al., 2025).

(o) The hydrogel soaked with moringa leaf extract to be used as the experimental treatment.

Gel-Based Growing System

(c) A cultivation system where plants grow in water-retaining polymer gels instead of soil (Ahmad, 2015).

(o) The method to be used in this study where seeds are grown in hydrogel.

Germination Percentage

(c) The proportion of seeds that successfully germinate under defined conditions (ISTA, 2020).

(o) The number of mung bean seeds that sprouted divided by the total seeds used.

Hydrogel

(c) A three-dimensional polymer network capable of absorbing and retaining large amounts of water (Ahmad, 2015)

(o) The gel medium used as a substitute for soil in growing mung bean seeds.

Mean Germination Time (MGT)

(c) represents the average time required for seeds to germinate and is used to evaluate the speed and uniformity of germination (Kulkarni et al., 2007).

(o) The average number of days it took for mung bean seeds to sprout.

Moringa *oleifera*

(c) A fast-growing multipurpose tree species widely cultivated in tropical regions (FAO, 2014).

(o) The plant whose leaves were collected and used to prepare the extract in this study.

Moringa Leaf Extract

(c) A plant extract obtained from Moringa oleifera leaves containing bioactive compounds (Alhusnan & Alkahtani, 2016)

(o) The aqueous solution infused into hydrogel as the treatment material.

Mung Bean (Vigna radiata)

(c) A leguminous crop commonly used in germination studies due to rapid growth (FAO, 2014)

(o) The test plant used in this study.

Root Length

(c) Measurement of root development used to assess seedling growth (ISTA, 2020).

(o) The measured length of the mung bean root in centimeters.

Seed Germination

(c) A crucial stage in the life cycle of plants, as it marks the beginning of their growth and development (Iqbal, 2023).

(o) The visible sprouting of mung bean seeds grown in hydrogel during the experimental period.

Seedling

(c) A young sporophyte developing out of a plant embryo from a seed (ISTA, 2020).

(o) A germinated mung bean used for measurement.

Seedling Vigor Index (SVI)

(c) An index combining germination percentage and seedling growth parameters (Abdul-Baki & Anderson, 1973).

(o) The value calculated using germination percentage and seedling length.

Shoot Length

(c) The measurement of the height of a seedling's above-ground portion (ISTA, 2020)

(o) The measured length of the mung bean shoot in centimeters.

Water Availability

(c) The accessible water supply influencing plant growth (FAO, 2014)

(o) The water provided by hydrogel during germination.

Water Retention

(c) The capacity of a medium to retain water for plant use (Ahmad, 2015)

(o) The capacity of hydrogel to store moisture for seed use.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents a review of related literature and studies that support the present research. The reviewed materials are grouped based on their relevance to hydrogels, moisture retention in gel-based systems, and the use of extracts infused into gels. These studies provide the scientific foundation for the use of hydrogels in soilless agriculture and highlight the need to explore extract-infused gels for improving moisture availability to seedlings.

2.1 Hydrogels

Hydrogels are water-absorbent polymer networks capable of retaining large amounts of moisture and releasing it gradually over time. Due to this property, they have been increasingly studied as materials for improving water availability in agriculture and soilless systems, particularly under conditions of irregular irrigation or limited water supply (Muhammad et al., 2025). In addition to their high water-absorption capacity, hydrogels possess elastic and porous structures that allow them to expand when hydrated and shrink when moisture is released. This reversible swelling behavior enables hydrogels to respond dynamically to environmental moisture conditions. Such characteristics make hydrogels suitable for use in controlled growing environments where maintaining consistent moisture levels is essential for seed germination and early seedling development. Their non-toxic and biodegradable nature further supports their application in sustainable agricultural practices.

Recent studies consistently show that hydrogels function as temporary water reservoirs. Muhammad et al. (2025) demonstrated that biodegradable agriculture hydrogels significantly improved moisture retention and reduced water loss through evaporation, thereby stabilizing water availability during early plant growth stages. Similarly, a study by Agbna & Zaidi (2025) found that hydrogels offer significant potential to enhance crop resilience, improve water use efficiency, and promote sustainable farming practices. Another important advantage of hydrogels is their ability to improve water-use efficiency by minimizing water loss due to leaching and evaporation. By holding water within their polymer networks, hydrogels reduce excessive drainage and allow plants to access moisture over extended periods. This property is particularly beneficial in soilless and gel-based systems, where water availability directly depends on the growing medium. As a result, hydrogels contribute to more stable growing conditions, especially during the sensitive germination stage.

Recent studies also show that hydrogels can function as controlled-release carriers for nutrients in agricultural systems. According to Singh et al. (2023), slow-release NPK fertilizer hydrogels improve nutrient availability while maintaining high moisture retention, supporting plant growth under limited water conditions. Similarly, Winarti et al. (2021) reported that incorporating NPK fertilizer into nanocellulose-based hydrogels altered swelling behavior and moisture-holding capacity, demonstrating that added substances can modify hydrogel properties.

While these studies confirm the effectiveness of hydrogels in conserving water, they primarily treat hydrogels as passive materials designed only to store and release moisture. Limited attention has been given to their potential role as functional matrices capable of carrying additional substances. A study by Kawee-ai (2025) states that the integration of natural extracts into gel systems has emerged as a transformative approach to enhance functional properties, including antioxidant, antimicrobial, and therapeutic effects. Although this study focuses on the biomedical purpose of extract-infused gel systems, it provides a clear gap that the purpose of extract-infused gel systems be used in agriculture.

Taken together, these findings indicate that hydrogels are effective in managing moisture availability but remain underexplored as multifunctional materials in soilless agriculture. This gap supports the investigation of extract-infused hydrogels, where moisture retention and extract delivery are combined to enhance seed germination in gel-based growing systems.

2.1.1 Extract Infusion on Hydrogels

The infusion of plant-based extracts into hydrogels has gained attention as a strategy to enhance the functionality of gel-based systems beyond moisture retention. Extracts derived from organic sources contain bioactive compounds that can influence biological activity and stress responses. When incorporated into hydrogels, these compounds can be stored within the gel matrix and released gradually alongside moisture. The effectiveness of extract infusion depends on the compatibility between the plant extract and the polymer matrix of the hydrogel. Proper incorporation allows the extract to be uniformly distributed within the gel structure, preventing rapid loss or uneven release. This controlled interaction ensures that bioactive compounds remain available over time, rather than being immediately washed away. Such controlled delivery systems are advantageous in agricultural settings where sustained exposure to growth-promoting substances is desired.

A study published in *Gels* (MDPI, 2024) demonstrated that hydrogels infused with natural extracts maintained their water-absorbing capacity while enabling controlled release of extract compounds. This dual function allowed the gel to act simultaneously as a moisture reservoir and a carrier medium. Similarly, research reported in *Carbohydrate Polymers* (2023) found that extract-loaded biodegradable hydrogels exhibited sustained release behavior, indicating their suitability for prolonged interaction with biological systems.

A recent review by Kawee-ai (2025) further supports the successful incorporation of plant-based extracts into hydrogels, showing that bioactive compounds such as polyphenols and flavonoids can be immobilized within gel matrices without significantly reducing water absorption capacity. These extract-infused hydrogels retained their moisture-holding function while enabling gradual release of bioactive substances, highlighting their dual functionality as both water reservoirs and delivery systems.

Despite these findings, most existing studies focus on biomedical, pharmaceutical, or material science applications of extract-infused hydrogels rather than agricultural or soilless cultivation contexts. For instance, Hossainpour et al. (2025) developed gelatin–sodium alginate hydrogels infused with *Onosma dichroanthum* Boiss. for wound dressing applications. However, the effects of extract-infused hydrogels on localized moisture availability at seed–medium contact points, particularly in gel-based growing systems, remain insufficiently examined. Furthermore, limited research has explored how extract infusion alters the moisture retention behavior of hydrogels themselves, highlighting a critical gap for agricultural applications.

In agricultural applications, controlled-release systems are particularly important during early growth stages when seeds and seedlings are highly sensitive to environmental conditions. The gradual release of plant extracts from hydrogels may help maintain a stable biochemical environment around the seed. This approach could reduce stress caused by sudden changes in moisture or nutrient availability. However, further investigation is needed to determine how extract infusion influences hydrogel behavior under germination conditions.

2.1.2 Hydrogel Moisture Retention

Hydrogels retain water through three-dimensional polymer networks that enable absorption, storage, and gradual release of moisture. Studies indicate that factors such as polymer type, concentration, and cross-linking density influence both swelling capacity and water-release rate, thereby determining how effectively the gel supplies moisture to plant roots or seed contact points (Muhammad et al., 2025; *Colloids and Surfaces B*, 2025).

Moisture retention in hydrogels plays a critical role in maintaining seed hydration during germination. Consistent moisture availability supports enzymatic activity and cellular expansion required for radicle emergence. Unlike traditional growing media that may dry out quickly, hydrogels maintain prolonged contact between water and seeds. This sustained hydration is especially beneficial for fast-germinating crops that require uninterrupted moisture supply.

In agricultural contexts, hydrogels act as temporary water reservoirs that buffer against fluctuations in moisture availability. Research has shown that biodegradable hydrogels can maintain consistent moisture around seedlings while reducing evaporation, supporting early growth even under irregular irrigation conditions (*Scientific Reports*, 2025). These findings highlight that hydrogels are not merely passive water stores but function as systems that modulate moisture delivery according to environmental conditions.

However, hydrogel behavior may change when plant extracts are incorporated into the gel matrix. Studies suggest that extract infusion can influence swelling behavior and water-release dynamics, potentially modifying moisture retention performance (*Journal of Agricultural and Food Chemistry*, 2016; Kawee-ai, 2025). Understanding these interactions is essential for designing hydrogels that provide both sustained hydration and controlled release of bioactive compounds.

In addition, plant extract incorporation may further modify polymer interactions, altering the swelling and deswelling behavior of the gel. Some bioactive compounds can interact with polymer chains, potentially affecting water uptake and release rates. These interactions may either enhance or slightly reduce moisture retention, depending on extract concentration and gel composition. Understanding these effects is essential for optimizing hydrogel formulations for agricultural use.

2.2 Moringa Leaf Extract

Moringa oleifera leaf extract has been widely studied for its growth-promoting potential due to its rich biochemical composition, which includes essential nutrients, amino acids, antioxidants, antifungal and naturally occurring phytohormones such as cytokinins and auxins (Fahey, 2005; Leone et al., 2015). These compounds regulate physiological processes including cell division, root initiation, and shoot development, making moringa leaf extract a promising natural growth enhancer in agriculture.

In addition to phytohormones, moringa leaf extract contains minerals such as calcium, potassium, and magnesium, which are essential for early plant development. These nutrients support root formation, enzyme activation, and cellular metabolism during germination. The natural origin of moringa extract also makes it an environmentally friendly alternative to synthetic growth enhancers. This supports its increasing use in sustainable and low-cost agricultural practices.

Several studies have reported that moringa leaf extract improves seed germination, seedling vigor, and early plant development. The presence of cytokinin compounds, particularly zeatin, has been associated with delayed leaf senescence and increased chlorophyll content, enhancing photosynthetic efficiency and overall plant performance (Howladar, 2014; Rady et al., 2015). Additionally, the antioxidant properties of moringa leaf extract help reduce oxidative stress, thereby improving plant tolerance to environmental challenges. According to Islam et al. (2022) moringa contains very high levels of zeatin compared to other plants. Zeatin is a phytohormone (cytokinin) that promotes cell division, shoot and root development, and delays aging in plant tissues.

Most existing studies apply moringa leaf extract through foliar spraying or soil-based methods. While these approaches demonstrate positive effects on crop performance, limited research has explored the use of moringa leaf extract in soilless or gel-based cultivation systems. Given the moisture-retention capabilities of hydrogels, combining moringa leaf extract with hydrogel-based systems presents a novel approach that integrates biological stimulation with physical water regulation.

Applying moringa leaf extract through gel-based systems may overcome several limitations associated with conventional application methods. Gel matrices can protect sensitive bioactive compounds from rapid degradation caused by light, heat, or microbial activity. Additionally, controlled release from hydrogels may provide continuous exposure to growth-promoting substances during critical early stages. This approach may result in more uniform and predictable germination outcomes.

In this system, the hydrogel may act as a carrier that allows gradual release of bioactive compounds from the moringa extract, potentially enhancing nutrient availability during early growth stages. However, empirical evidence on the effectiveness of moringa-infused hydrogels remains limited, particularly for fast-germinating crops such as mung beans. This gap supports the need to evaluate the combined effects of moringa leaf extract and hydrogels on seed germination.

2.3 Seed Germination

Seed germination is a critical stage in plant development, as it determines early seedling establishment and overall growth potential. Germination begins with water uptake, which activates metabolic processes leading to radicle emergence. Insufficient or inconsistent moisture during this phase can delay germination and reduce seedling vigor (*Journal of Plant Nutrition*, 2008; *Cogent Food & Agriculture*, 2023).

The speed and uniformity of germination are important indicators of seed quality and growing conditions. Rapid and synchronized germination allows seedlings to establish evenly, reducing competition and increasing survival rates. Environmental factors such as temperature, oxygen availability, and moisture interact to influence germination success. Among these factors, moisture availability is considered one of the most critical determinants.

In soilless systems, moisture availability is governed by the properties of the growing medium rather than soil particles. Gel-based systems, including hydrogels, have been shown to provide stable moisture at the seed–medium interface. Studies indicate that hydrogels improve moisture retention and reduce water stress during early growth, which is particularly important for fast-germinating crops (Muhammad et al., 2025; *Scientific Reports*, 2025).

Uniform moisture delivery has been associated with higher germination rates and improved early seedling vigor. Gel-based media maintain close contact between seeds and hydrated matrices, promoting rapid imbibition while minimizing dehydration stress (*Journal of Experimental Agriculture International*, n.d.). Mung bean (*Vigna radiata*) is commonly used in germination studies due to its rapid and uniform germination response, allowing clear observation of treatment effects.

Using mung beans as a test crop allows researchers to observe treatment effects within a short period of time. Their sensitivity to moisture variations makes them suitable for evaluating the performance of hydrogel-based growing systems. Differences in germination rate and early growth can be easily measured, providing reliable data for comparative analysis. This makes mung beans an appropriate model organism for this study.

Despite evidence supporting hydrogel use, most studies examine hydrogels as standalone materials. Few have investigated how modifying hydrogels through plant extract infusion influences germination outcomes. This gap highlights the need to explore whether extract-infused hydrogels can enhance germination beyond moisture retention alone.

2.4 Aqueous Extraction on Moringa Leaf

Previous studies evaluating hydrogels in agricultural and soilless systems commonly assessed moisture retention and early plant growth under controlled conditions. These studies typically compared hydrogel-treated systems with untreated controls to determine differences in water availability and seedling performance (Muhammad et al., 2025; *Scientific Reports*, 2025).

Controlled experimental designs are essential in evaluating the effectiveness of hydrogel-based systems. Variables such as temperature, light exposure, and seed type are typically standardized to ensure reliable results. Replication is also commonly used to reduce experimental error and improve data validity. These methodological practices strengthen the accuracy and credibility of research findings.

Studies involving plant extracts frequently measured germination rate, seedling height, and early vigor. Extracts were commonly prepared using aqueous extraction methods to preserve bioactive compounds and applied to seeds or growth media (*Journal of Plant Nutrition*, 2008; *Cogent Food & Agriculture*, 2023).

Al Husnan & Alkahtani (2016) air dried fresh moringa leaves at room temperature to constant weight. The dried leaves we're then grounded in powders. They used aqueous extraction instead of ethanol extraction.

Research on extract-infused hydrogels, largely conducted in biomedical and material science fields, focused on controlled release behavior and material stability rather than plant growth outcomes (Kawee-ai, 2025). Nevertheless, these methodologies demonstrate that extract incorporation enables gradual release alongside moisture, a principle applicable to agricultural systems.

2.5 Research Gaps

Although hydrogels are established as effective moisture-retention materials, most studies treat them as passive water-holding media. Limited research has examined how modifying hydrogels with plant extracts influences seed germination outcomes. Similarly, moringa leaf extract is widely studied as a growth enhancer, but its application through controlled-release systems such as hydrogels remains underexplored.

Recent studies demonstrate a growing interest in the use of hydrogels and plant extracts as sustainable solutions for improving plant growth and resource efficiency. Research on agricultural hydrogels consistently agrees that these materials enhance water retention, regulate moisture availability, and support plant establishment, particularly under limited water conditions (Muhammad et al., 2025; Scientific Reports, 2025). These studies emphasize the physical role of hydrogels as water reservoirs, reducing irrigation frequency and improving seedling survival. However, most agricultural hydrogel studies focus primarily on moisture regulation and do not incorporate bioactive components that could further enhance plant physiological performance.

On the other hand, extensive research on plant extracts—particularly Moringa oleifera leaf extract—has established their effectiveness in improving seed germination, plant growth, and stress tolerance. Studies consistently report that moringa leaf extract contains growth-promoting compounds such as cytokinins, antioxidants, and essential nutrients that positively influence early plant development (Fahey, 2005; Leone et al., 2015; Rady et al., 2015). These studies largely agree on the biological benefits of moringa extracts; however, they are commonly applied through foliar spraying or soil drenching, methods that may result in uneven distribution, rapid degradation, or nutrient loss.

Meanwhile, studies on extract-infused hydrogels are predominantly situated in biomedical and pharmaceutical contexts. These studies demonstrate that hydrogels are effective carriers for controlled and sustained release of bioactive compounds, maintaining stability and prolonging functionality over time (Carbohydrate Polymers, 2023; Journal of Agricultural and Food Chemistry, 2016; Gels, 2024). Although these findings confirm the technical feasibility of embedding natural extracts into hydrogel matrices, their application to plant systems—particularly during seed germination—remains limited.

In addition, many existing studies rely on advanced laboratory equipment and materials that may not be accessible in school-based research settings. There is limited literature demonstrating the feasibility of extract-infused hydrogels using simple preparation methods and low-cost materials. Addressing this limitation is important for expanding the practical application of hydrogel technology in educational and community-based agricultural projects.

Furthermore, existing research on extract-infused hydrogels primarily focuses on biomedical applications, leaving agricultural and soilless cultivation contexts insufficiently examined. There is limited evidence on how extract infusion affects hydrogel moisture-retention behavior and localized water availability during germination.

Therefore, a clear research gap exists regarding the combined use of moringa leaf extract–infused hydrogels in gel-based growing systems. This study addresses this gap by investigating their effects on seed germination, contributing new insight into multifunctional hydrogel applications in soilless agriculture.

Synthesis of the State of the Art

Agbna and Zaidi (2025) reviewed the current performance of agricultural hydrogels and emphasized their role in improving soil water retention, enhancing plant tolerance to water stress, and supporting sustainable crop production. Their review highlights that hydrogels act as water reservoirs, gradually releasing moisture to plant roots during drought conditions. This finding is strongly supported by Muhammad et al. (2025), who similarly concluded that agricultural hydrogels significantly improve moisture retention and reduce irrigation frequency, thereby increasing crop resilience under limited water availability. Scientific Reports (2025) further agrees with these conclusions by demonstrating that biodegradable hydrogels regulate moisture in plant systems and maintain soil hydration over extended periods, reinforcing the consensus that hydrogels are effective tools for mitigating water stress in agriculture.

The importance of polymer structure in hydrogel performance is emphasized by Colloids and Surfaces B: Biointerfaces (2025), which showed that polymer network density directly influences water absorption and retention behavior. This agrees with the findings of Singh et al. (2023), who reviewed slow-release NPK fertilizer hydrogels and noted that crosslinking density and polymer composition are critical factors controlling nutrient and water release. Winarti et al. (2021) also support this perspective, showing that incorporating NPK fertilizer into nanocellulose-based hydrogels alters swelling behavior and mechanical properties, confirming that hydrogel formulation determines functional efficiency. Together, these studies agree that optimizing polymer structure is essential for achieving controlled release and effective agricultural performance.

Several studies focus on the integration of plant extracts into hydrogel systems. The Carbohydrate Polymers (2023) study demonstrated that extract-loaded biodegradable hydrogels exhibit controlled release behavior, ensuring sustained availability of bioactive compounds. This finding is supported by the Journal of Agricultural and Food Chemistry (2016), which showed that polymeric gel matrices can successfully encapsulate plant extracts while preserving their biological activity. Kawee-ai (2025) further agrees by reporting that plant extract–embedded hydrogels maintain biological functionality, including antioxidant and antimicrobial properties, suggesting that hydrogels are effective carriers for bioactive plant compounds. Similarly, Hossainpour et al. (2025) demonstrated that gelatin–sodium alginate hydrogels infused with plant extracts enable controlled release, reinforcing the shared conclusion that hydrogel systems can enhance the stability and delivery efficiency of plant-derived bioactives.

The use of natural extracts, particularly Moringa oleifera, as plant growth enhancers is consistently supported across multiple studies. Fahey (2005) provided foundational evidence that moringa leaves and seeds contain high levels of nutrients, antioxidants, and bioactive compounds, which explains their biological effectiveness. Leone et al. (2015) and Islam et al. (2022) expanded on this by confirming the nutritional richness of moringa and its functional properties, indirectly supporting its agricultural applications. Howladar (2014) directly demonstrated that moringa leaf extract mitigates salt stress in plants by improving growth and physiological performance. This finding is strongly supported by Rady et al. (2015), who showed that moringa leaf extract improves plant growth and productivity under stress conditions, indicating agreement that moringa functions as a biostimulant and stress alleviator in plants. Al Husnan & Alkahtani (2016) provided methods in extracting moringa leaf by using aqueous extraction methods by grounding the leaves and distilling it with water, which is verified to extract the moringa leaf's bioactive composition.

Seed germination and early seedling development are identified as critical stages influenced by moisture availability and growing media. The Journal of Plant Nutrition (2008) established that seed moisture uptake directly affects germination rate and early seedling growth, highlighting the importance of consistent water availability. This agrees with Cogent Food & Agriculture (2023), which found that moisture availability and growing media significantly influence germination percentage and seedling vigor. The Journal of Experimental Agriculture International (n.d.) similarly reported that growing media affect germination behavior in leguminous crops, reinforcing the consensus that both moisture regulation and substrate conditions are key determinants of successful germination. These findings collectively support the application of hydrogels as moisture-regulating agents during early plant development.

When viewed together, the studies consistently agree that hydrogels improve water retention, regulate moisture availability, and support plant growth, particularly under stress conditions. Multiple studies also converge on the idea that embedding bioactive plant extracts, such as moringa, into hydrogels enhances their effectiveness by enabling controlled release and prolonged biological activity. While individual studies differ in materials used, plant species tested, and application methods, there is strong overall agreement that biodegradable, extract-infused hydrogels represent a state-of-the-art approach for improving seed germination, seedling vigor, and plant resilience in sustainable agriculture.

Conclusion

Our study aims to innovate the existing hydrogel gel-based system by infusing it with moringa leaf extracts and addressing the issues with irregular rainfall, irrigation, and dry seasons. The study is feasible because of the abundance of *Moringa Oleifera* leaf and extraction methods have been done. The study is unique because most studies about extract-infused hydrogels focuses on biomedical reasons and limited studies focuses on agricultural reasons. The need for studying can provide scientific evidence and data on how these extract-infused hydrogels affect seed germination or fungal activities show which treatment is most optimal for seed germination.

CHAPTER 3
METHODOLOGY

This chapter focuses on the techniques and research methodologies that were chosen by the researchers to address the issues. Specifically, the data collection and preparation, the application of treatment, and the data analysis.

CAUSAL DIAGRAM

CONFOUNDING VARIABLE

- Type and size of container
- Number of replicates
- Age of mung bean
- Size of mung bean
- Duration of experiment
- Concentration of Moringa Extract
- Volume of Solution
- Extraction Method
- Temperature

INDEPENDENT VARIABLE

- *Moringa oleifera* leaf extract-infused hydrogels
- *Moringa oleifera* hydrogel concentration

DEPENDENT VARIABLE

- Seed Germination
- Germination Percentage
- Mean Germination Time
- Seedling Vigor Index
- Shoot Length
- Root Length
- Physical property
- Chemical Property

EXTRANEOUS VARIABLES

- Environmental Factors

RESEARCH DESIGN

This study entitled "Effect of Moringa (*Moringa Oleifera*) Leaf Extract-Infused Hydrogels on Seed Germination of Mung Beans (*Vigna radiata*) in a Gel-Based Growing System." The experimental units in the study were set up using the Completely Randomized Design (CRD) with 3 treatments each with 10 replicates.

Table 1. Experimental Layout

CT1	MB9	MB5	MB29	MB11	MB17	MB30	MB21	MB8	MB19	MB3
T1	MB10	MB6	MB12	MB15	MB16	MB13	MB23	MB1	MB26	MB20
T2	MB14	MB7	MB4	MB22	MB18	MB25	MB24	MB2	MB27	MB28

Wherein:

Control Treatment 1 (CT1) – 0% leaf extract concentration

Treatment 1 (T1) – 5% leaf extract concentration

Treatment 2 (T2) – 15% leaf extract concentration

MB – Mung Beans (*Vigna Radiata*)

The Researchers used the study of Al-Husnan & Alkahtani entitled "Impact of *Moringa* Aqueous extract on pathogenic bacteria and fungi *in vitro*" as the basis of their methodology. The researchers also used the study of Benramadan entitled "Effect of Different concentration of the Aqueous leaf extract of Moringa olifeira on seed germination and early growth of *Lycopersicon esculentum mill*" as the basis for the concentration of moringa *oleifera* extract.

The researchers will use the studies of both Al-Husnan & Alkahtani and Paramita as the basis of the drying method. According to Paramita (2023) oven drying at 40°C showed the best result in moringa *oleifera* drying. Oven drying is the priority drying method, but in cases where a laboratory oven is not available, the researchers shall use the air drying method.

EQUIPMENT AND TOOLS

The tools and equipment used in this study were rubber glove and a machete for the collection of *Moringa oleifera leaves*. Dried moringa leaves, mortar and pestle, weighing scale, conical flask, beaker, filter paper, funnel, stirring rod, shaker incubator, graduated cylinder, timer, distilled water, surgical masks, and plastic gloves for extraction. Laboratory oven for drying leaves. Petri dishes, hydrogel mixture, and water for infusion. Labels and marker pens for experimental setup. Ruler for measurement.

METHODS AND PROCEDURES

This experiment was conducted In 10 phases: (1) Preparation of Materials, (2) Collection and Identification of Moringa Leaves, (3) Drying of Leaves, (4) Grinding and Powder Preparation, (5) Aqueous Extraction, (6) Filtration and Preparation of Extract Concentrations, (7) Hydrogel Preparation and Infusion, (8) Application of Treatments, (9) Observation and Data Collection, (10) Data Analysis

a. Preparation of Materials

The mung bean (*Vigna Radiata*) seeds used in the experiment were bought from shopee. Commercial agricultural hydrogel was purchased in shopee and prepared as the soilless growing medium. Laboratory equipments were borrowed from the Teaching Resource Center (TRC) of Nabua National High School and other available laboratory facilities. In case materials aren't available, alternative equipment shall be used.

b. Collection and Identification of Moringa Leaves

Fresh leaves of *Moringa oleifera* were collected from mature plants in the local area. Species identification was conducted based on morphological characteristics described by FAO (2014).

c. Drying of Leaves

Collected leaves were washed with distilled water and dried using a laboratory oven at approximately 40°C until dry to reduce moisture content while preserving bioactive compounds (Paramita, 2023). If laboratory oven is not available, air-drying method shall be used. The fresh leaves were air dried at room temperature to constant weight for 10 days.

d. Grinding and Powder Preparation

Dried leaves were ground using a mortar and pestle to obtain fine powder suitable for extraction.

e. Aqueous Extraction

Approximately 25g of powdered leaves were soaked in 100mL distilled water in a conical flask or erlenmeyer flask. The mixture was allowed to macerate to facilitate diffusion of bioactive compounds, including naturally occurring cytokinins such as zeatin, into the solvent. The flask was shaken at approximately 120rpm for 30 minutes.

f. Filtration and Preparation of Extract

The maceration mixture was filtered using filter paper and a funnel to separate solid residues from the liquid extract. The filtrate was collected as crude aqueous moringa extract.

g. Hydrogel Preparation and Infusion

Plantita’s Choice hydrogel was hydrated according to the manufacturer's instructions. The hydrogels were allowed to swell in treatment solutions to enable absorption of bioactive compounds through diffusion.

h. Application of Treatments

The treatment concentrations were prepared as percentage volume per volume (% v/v) solutions by mixing measured volumes of moringa leaf extract with distilled water. For the 5% concentration, 1mL of leaf extract was combined with 19mL distilled water to obtain a total volume of 20mL solution. For the 15% concentration, 3mL of leaf extract was combined with 17mL to obtain a total volume of 20mL solution. The 0% concentration served as the control as consisted of distilled water only without moringa extract. 30 petri dishes were used in total, 1 for each replicate. Each prepared solution as used to soak the hydrated hydrogel to allow infusion prior to seed placement. Mung bean seeds were placed on the hydrogel surface in petri dishes.

i. Observation and Data Collection

Seeds were monitored daily. Germination percentage, mean germination time, shoot length, root length, and seedling vigor index were measured by the amount of germination over total seeds, time it took to germinate, ruler, and a formula respectively.

- Germination Percentage – refers to the proportion of seeds that successfully germinate relative to the total number of seeds tested under specific experimental conditions (Kulkarni et al., 2007)

Germination Percentage (GP):

$$GP = \frac{\text{Number of Total Germinated Seeds}}{\text{Total Number of Seeds Tested}} \times 100$$

- Mean Germination Time – represents the average time required for seeds to germinate and is used to evaluate the speed and uniformity of germination (Kulkarni et al., 2007).

- Shoot Length – refers to the measurement of the aerial portion of the seedling from the base to the shoot tip, used to evaluate early vegetative growth (Kulkarni et al., 2007).

- Root Length – measurement of the primary root from the seed base to the root tup and is used to assess early root system development (Sunil Kamar et al., 2021)

- Seedling Vigor Index – is an integrated parameter combining germination percentage and seedling length to evaluate overall seedling strength and establishment potential (Sunil Kamar et al., 2021; Abdul-Baki & Anderson, 1973).

$$Seedling\ Vigor\ Index = \ Germination\,\% \ \times (Shoot\ length\ \bar{x} + Root\ length\ \bar{x})$$

j. Data Analysis

In order to determine if there were any significant difference among the treatments: the positive control and the 5% and 15% leaf extract, in terms of the germination percentage, the One-Way Anova was used for statistical analysis. If significant differences were computed between the treatments, a post hoc test shall be done.

RESEARCH TIMELINE																																
MONTH	CODE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
FEBRUARY	A																															
MARCH	B																															
	C																															
	D																															
	E																															
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JUNE	G																															
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	M																															
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	P																															
	Q																															

Legend:

A- Process of making the Concept Note.

B- Finalizing and submitting the Concept Note.

C- Requesting for a Consultation.

D- Revision of research title, rationale, problem, expected outcomes, hypothesis.

E- Making the Research Plan.

F- Finalizing the Research Plan.

G- Consultation on Research Methodology Adviser.

H. Preparation of Materials.

I. Collection of Plant Materials

J. Identification of Species

K. Drying Process (Black – Priority, Red – Alternative)

L. Maceration and Aqueous Extraction.

M. Filtration and Extract Preparation.

N. Hydrogel Preparation and Infusion.

O. Application of Treatments.

P. Observation and Gathering of Data.

Q. Data Analysis and Interpretation.

TASK CODE	TASK DESCRIPTION	IMMEDIATELY PRECEEDING TASKS	ESTIMATED DURATION (in days)	OBSERVABLE INDICATORS*
A	Process of making the Concept Note	B	Feb 7 to March 29 (23 days)	Logbook
B	Finalizing the Concept Note	C	March 2 to 3 (2 days)	Logbook
C	Requesting for a Consultation	--	March 4 & 7	Logbook
D	Revision of rationale, problem, expected outcome, and hypothesis	E	March 6 to 20 (15 days)	Research plan and logbook
E	Making the Research Plan	F	March 9 to 20 (12 days)	Research plan and logbook
F	Finalizing the Research Plan	--	March 20 to 21 (2 days)	Research plan and logbook
G	Consultation on Research Methodology Adviser	H		Logbook
H	Preparation of Materials	G	June 1 to June 2 (2 days)	Complete Laboratory tools and materials ready Logbook
I	Collection of Plant Materials	H	June 1 to June 2 (2 days)	Plant Materials are collected Logbook
J	Identification of Species	I	June 3	Morphological characteristics confirmed
K	Drying Process	J	June 5 (June 5 – June 15 if alternate drying is used)	Leaves visibily dried and brittle

L	Maceration and Aqueous Extraction	K	June 15	Leaf powder soaked in distilled water
M	Filtration and Extract Preparation	L	June 15	Extract solution prepared and filtered extract obtained
N	Hydrogel Preparation and Infusion	M	June 19	Hydrogel medium ready
O	Application of Treatments	N	June 22	Seeds placed in treatment setup
P	Observation and Data Gathering	O	June 22 – June 29 (7 days)	Germination and growth measurements recorded
Q	Data Analysis and Interpretation	P	June 30 to July 2 (3 days)	Data analyzed and computed

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RISK ASSESSMENT AND MANAGAMENT

	RISKS	RISKS MANAGEMENT
Physical Hazard	• Cuts or Injury during moringa leaf collection • Burns from oven drying • Glassware breakage (Beaker, petri dish) • Minor injuries from tools (Mortar and Pestle)	• Use gloves and protective equipment • Handle hot materials with heat-resistant protection • Proper handling and storage • Work under supervision
Chemical Hazard	• Exposure to plant extracts causing skin irritation • Contamination during extraction	• Use gloves and masks • Avoid direct contact with extracts • Use clean and sterilized equipment
Biological Hazard	• Mold or microbial contamination during germination • Exposure to plant material allergens	• Maintain clean working environment • Monitor moisture levels • Dispose plant waste properly

FORMS

Student Checklist(1A)

Approval Form(1B)

Risk Assessment Form(3)

Adult Sponsor Form(1)

Hazardous Chemicals, Activities or Devices Risk Assessment(6A)

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